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Air nanobubble simultaneously enhances hydrolysis and methane yield of sludge temperature phased-anaerobic digestion☆

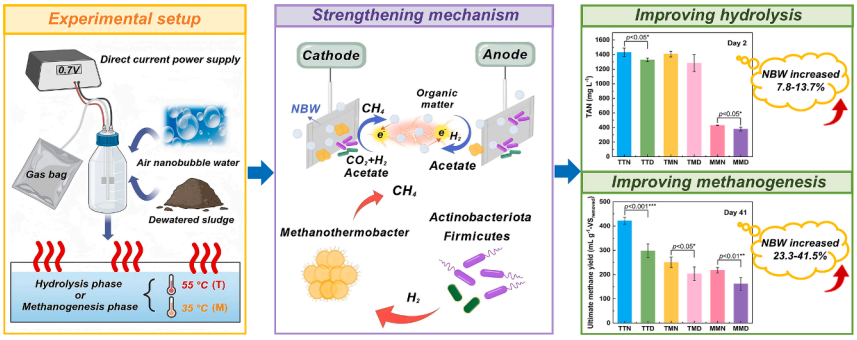
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HIGHLIGHTS

- NBW increased TAN by 7.8%-13.7% in the hydrolysis phase.
- NBW increased ultimate methane yield by 23.3%-41.5%.
- NBW improved the diversity and richness of microorganisms in hydrolysis.
- Thermophilic conditions facilitate low VS/TS sludge anaerobic digestion.
- The combination of NBW and MEC achieved efficient anaerobic digestion.

GRAPHICAL ABSTRACT



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ABSTRACT

Nanobubble water (NBW) or temperature-phased anaerobic digestion assisted by microbial electrolysis cell (MEC-TPAD) can promote sludge hydrolysis and methanogenesis. However, the role of the combined application of NBW and MEC-TPAD in terms of anaerobic performance and related microbial properties remains unclear. This study investigated the impact of Air-NBW on hydrolysis and methanogenesis of dewatered sludge MEC-TPAD. Under different temperatures, NBW increased ammonia nitrogen by 7.8%-13.7% in the hydrolysis phase and ultimate methane yield by 23.3%-41.5%. NBW can significantly promote hydrolysis under mesophilic-mesophilic conditions, while it can promote substantially methanogenesis under thermophilic-thermophilic conditions. Moreover, NBW increased the diversity and richness of microorganisms in hydrolysis. As to bacteria, NBW increased the relative abundance (RA) of *Firmicutes* but decreased the RA of *Proteobacteria*. As to archaea, NBW increased the RA of *Methanosarcina* in hydrolysis but decreased it in methanogenesis. NBW synchronized with MEC-TPAD improved hydrolysis and methanogenesis of the dewatered sludge digestion process.

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1. Introduction

Dewatered sludge (DWS) has become a critical issue in municipal wastewater treatment plants because the presence of harmful substances, heavy metals, and pathogenic microorganisms in DWS increases the potential risks to humans and the environment (Li et al., 2018). Temperature-phased anaerobic digestion (TPAD) has been extensively employed for the treatment of dewatered sludge (Wang et al., 2018). Thermophilic conditions effectively destroy pathogens and significantly enhance volatile solids (VS) removal in the hydrolysis phase, whereas mesophilic conditions mitigate volatile fatty acid (VFA) accumulation in the methanogenesis phase (Sillero et al., 2023). However, the excessive ammonia produced during thermophilic hydrolysis in TPAD may inhibit the activity of methanogenesis microorganisms (Sillero et al., 2023).

Coupling microbial electrolysis cell (MEC) with TPAD has been recognized as a promising technique to alleviate the inhibition of methanogen activity (Yuan et al., 2022). MEC modified the microbial community by enriching electroactive microorganisms and methanogens, which led to increased methane yield (Kanellos et al., 2024). MEC can also overcome thermodynamic limitations to accelerate the degradation of organic matter by applying voltage (Wang et al., 2022a). The intermittent MEC-TPAD increased ultimate methane yield (UMY) in the DWS by 118.9 % compared with the control group (Peng et al., 2024). However, the research of Peng et al. (2024) had a low total solids (TS) with 12.8 % and a high VS/TS with 63.0 % of DWS, which was conducive to the degradation of organic matter. Furthermore, distilled water was added prior to the digestion process to reduce the TS to 8.2 %, which improved mass transfer efficiency.

In actual circumstances, the TS of DWS is generally as high as about 20 % (Wang et al., 2024a). Excessively high TS will reduce the fluidity of digestate and affect the transfer efficiency of nutrients to microorganisms (Abid et al., 2021). Moreover, due to improper sewage treatment and low sediment interception rate, DWS often exhibits low organic matter content (VS/TS $<$ 50 %) (Shi et al., 2023). These factors are detrimental to the hydrolysis of sludge. In the case of high TS and low VS/TS, the hydrolysis of MEC-TPAD may still be insufficient (Shi et al., 2023; Wang et al., 2022a). Therefore, it is necessary to add a reasonable medium on the basis of MEC-TPAD to reduce the solid content and improve the hydrolysis performance.

Nanobubble water (NBW) is a water body containing nanobubbles with a diameter $<$ 1 μm and has the advantages of high stability, gas solubility, and negative zeta potential compared with deionized water (Pei et al., 2024). The high mobility of NBW facilitates the reduction of sludge viscosity and enhances nutrient transfer to microorganisms (Chuenchart et al., 2021). Furthermore, NBW enhanced the degradation of the complex substrate by releasing reactive oxygen species while stimulating enzyme activity to improve microbial community structure (Fan et al., 2021; Zhu et al., 2024). Additionally, O_2-NBW and Air-NBW enable the creation of micro-oxygen conditions for facultative microorganisms to promote substrate hydrolysis, thereby improving hydrolysis performance (Pei et al., 2024). NBW can stimulate the activity of methanogens, thereby increasing methane yield (Hou et al., 2024). Air-NBW increased methane yield from sludge by 23.0 % (Hou et al., 2021). Adding CO_2, Air, He, and N_2 to NBW improved the methane yield of DWS by 32.0 %, 27.9 %, 16.0 %, and 9.2 %, respectively (Wang et al., 2024a). The potential synergistic effects of NBW and MEC-TPAD on hydrolysis and methanogenesis warrant further investigation. Currently, there is limited research on the combined application of NBW and MEC, and the effects of NBW on the digestion performance and microbial community structure of MEC-TPAD remain largely unknown.

In this study, a low-cost and readily available Air-NBW coupled with intermittent MEC-TPAD was selected to (1) explore the effects of the new AD system on the hydrolysis efficiency, substrate degradation and methane yield of DWS; (2) 16S rRNA gene sequencing revealed the alteration of bacterial and archaeal communities by Air-NBW under different temperature conditions to better understand the metabolic mechanism of microorganisms in AD. This research contributes to developing a novel strategy for the combined application of NBW and MEC as an anaerobic treatment of DWS to increase renewable energy production.

2. Materials and methods

2.1. Materials

Thickened sludge and DWS were acquired from a wastewater treatment plant in Lanzhou utilizing an oxidation ditch process. Seed sludge was obtained by incubating thickened sludge under thermophilic and mesophilic conditions. The properties of DWS and different seed sludge were determined (**see Supplementary Material**). DWS (VS/TS $<$ 50.0 %) was the low organic matter content sludge (Shi et al., 2023). The design of the intermittent MEC anaerobic digesters was described by Peng et al. (2024). Air-NBW was prepared as described in Wang et al. (2024a).

2.2. Experimental process

At the beginning of the experiment, thermophilic or mesophilic seed sludge was added as an inoculum to DWS with Air-NBW or deionized water. The mass ratio of seed sludge, DWS, and Air-NBW or deionized water = 1: 5: 5, totaling 880 g, was added to each digester. Nitrogen was utilized to purge for 2–4 min to ensure an anaerobic environment in the digesters.

To determine the optimal temperatures for the hydrolysis and methanogenic phases in AD, respectively, the digesters were divided into three temperature conditions: T-T (sustained thermophilic conditions), M−M (sustained mesophilic conditions) and T-M (thermophilic + mesophilic conditions). Previous studies have found that the residence time of the hydrolysis phase was generally 2 ± 1 days (Ge et al., 2010; Mehari and Chang, 2022), so the T-M digesters chose to change the temperature after 2 days. The hydrolysis phase was set to be 0–2 days and the methanogenesis phase to be 3–41 days. After 2 days of the experiment, thermophilic seed sludge (80 g) was added to T-T digesters and mesophilic seed sludge (80 g) to T-M and M−M digesters. Depending on the addition of NBW or deionized water, the three temperature conditions were labeled TTN, TTD, TMN, TMD, MMN, and MMD. Digesters from different groups were placed in the corresponding water baths at different times (thermophilic for 55 ± 1 °C and mesophilic for 35 ± 1 °C). All groups were treated with MEC, and three parallel experiments were performed in each group.

2.3. Methods of analysis

TS, VS, pH, total ammonia nitrogen (TAN), and soluble chemical oxygen demand (SCOD) were measured in the same way as in APHA (1998). Free ammonia nitrogen (FAN) was analyzed utilizing standard formulas related to temperature, TAN, and pH (Hansen et al., 1998). A nanoparticle size analyzer was used to analyze sludge particle size (Malvern, Nano ZS90, UK). Methane content and VFA were assessed utilizing a gas chromatograph (Rui Hong, SP-7890Plus, China) equipped with a flame ionization detector and an organic acid analysis column (30 m × 320 μm × 0.5 μm). The set parameters of the column and detector were consistent with those used by Peng et al. (2024). A wet-type gas flow meter was used to analyze the biogas volume (Alpha, LML-2, China). A Spectro fluorophotometer was utilized to gain three-dimensional fluorescence excitation and emission matrix (3D-EEM) scans (Hitachi, F4600, Japan).

DNA was taken from sludge samples and then subjected to Illumina sequencing. Specific primer sequences 341F (CCTACGGGNGGCWGCAG) and 806R (GGACTACHVGGGTATCTAAT) were utilized to amplify the V3-V4 regions of bacteria. Specific primer sequences Arch519F (CAGCMGCCGCGGTAA) and Arch915R (GTGCTCCCCCGCCAATTCCT)

were utilized to amplify the V4-V5 regions of archaea. The sequenced microbial data have been included online (https://www.ncbi.nlm.nih.gov). PRJNA1151961 and PRJNA1151965 were the accession number.

2.4. Statistical analysis

Results Data were presented as the mean ± standard deviation of data from three parallel trials, and one-way analysis of variance (ANOVA) was performed using SPSS version 27.0. Statistical significance was set at $p < 0.05$. The Duncan's test and principal component analysis (PCA) were also performed using SPSS. The modified Gompertz formula was utilized to fit the methane yield (Zhang et al., 2021). The humification index (HIX) was calculated using the emission intensity region at 254 nm Ex (linear interpolation) (Guan et al., 2024). Figures were drawn using Origin 2018 (OriginLab, Massachusetts, USA).

3. Results and discussion

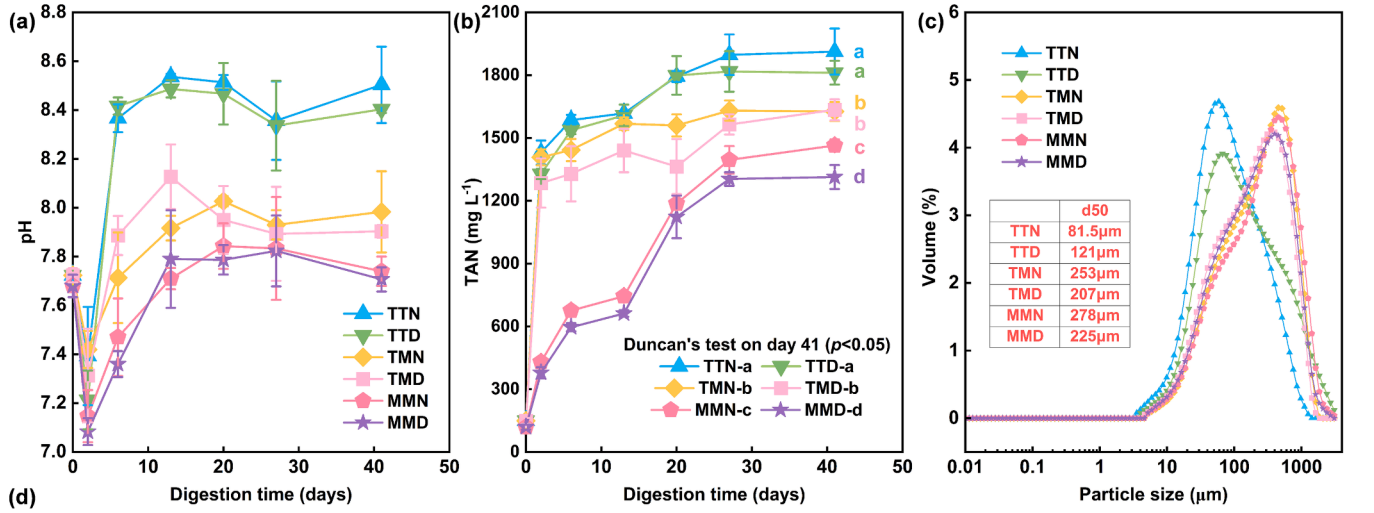
3.1. pH, TAN, sludge particle size, and VFA

The pH of all groups ranged from 7.0 to 8.5 (Fig. 1**a**), which was in line with the acceptable range for anaerobic microbial growth and metabolism in AD (Zhang et al., 2021). TAN can reflect the AD system's hydrolysis effects in the sludge (Wang et al., 2023). Methanogen activity may be inhibited when TAN exceeds 1500 mg L^{-1} (Wang et al., 2024a). On day 2, the TAN of the hydrolysis phase in T-M groups was increased by 227.1 %-239.6 % ($p < 0.001$) compared with that in M−M groups (Fig. 1**d**). In the hydrolysis phase, TAN in the NBW-added groups was improved by 7.8 %-13.7 % compared with the control groups. Among them, TAN was significantly increased in the T-T and M−M groups ($p < 0.05$), with the highest rate of increase observed in the M−M groups. NBW could promote the hydrolysis ability of the AD system, with the most pronounced effects observed under mesophilic conditions. On day 41, the TAN of the groups was T-T > T-M > M−M in descending order. Duncan's test based on day 41 sample data showed no correlation between the TAN of the three types of temperature conditions, which were significantly different (Fig. 1**b**). However, NBW did not produce a significant difference in TAN between groups except for the M−M groups.

The hydrolysis performance of AD under thermophilic conditions is superior to that under mesophilic conditions (Liu et al., 2013). Elevated temperatures are conducive to the improvement of hydrolysis performance. However, they also impact microbial diversity. Compared with thermophilic conditions, the types of hydrolytic bacteria under mesophilic conditions are more abundant (Chen et al., 2021a), and NBW could promote the activity of more types of hydrolytic bacteria to improve the hydrolysis performance. Consequently, NBW increased TAN more aggressively under mesophilic conditions. Moreover, temperature conditions significantly influenced TAN more than NBW. Different NBW increased TAN during AD by 6.4 %-10.1 % (Hou et al., 2022; Wang et al., 2024a). Compared with mesophilic conditions, thermophilic conditions during AD increased TAN by 59.0 %-173.7 % (Peng et al., 2024; Wang et al., 2023). These results suggested that temperature conditions are a more critical factor influencing AD.

FAN can penetrate the cell membrane of methanogens, making methanogens toxic (Zhang et al., 2021). The FAN in the T-T groups was significantly higher than in the T-M and M−M groups (**see Supplementary Material**). The activity of acetate-nutrient methanogens may be inhibited when FAN exceeds 588 mg L^{-1} (Moerland et al., 2021). From day 6 to day 41, the acetate-nutrient methanogens in the T-T digesters may be inhibited. However, hydrogenotrophic methanogens were more tolerant to high ammonia environments, and the activity of hydrogenotrophic methanogens could not recover from ammonia stress when FAN exceeds 900 mg L^{-1} (Wang et al., 2022b).

The smaller sludge particle size means the AD process was carried out more thoroughly (Guo et al., 2021; Liang et al., 2022). Under thermophilic conditions, the sludge particle size of the NBW-added group was smaller than that of the control (Fig. 1**c**). Conversely, under mesophilic conditions, the sludge particle size of the groups with NBW



Role of different treatments	Test	Control	TAN enhancement	Role of different treatments	Test	Control	TAN enhancement
Temperature difference (day 2)	TMN	MMN	227.1%***	Temperature difference (day 41)	TTN	TMN	17.6%*
	MD	MMD	239.6%***		TTD	TMD	10.8%*
NBW difference (day 2)	TTN	TTD	7.8%*	NBW difference (day 41)	TTN	TTD	5.5%
	TMN	TMD	9.5%		TMN	TMD	-0.4%
	MMN	MMD	13.7%*		MMN	MMD	11.5%*

Fig. 1. The (a) pH, (b) total ammonia nitrogen, and (d) total ammonia nitrogen enhancement ratio with different treatments during anaerobic digestion, (c) sludge particle size on day 41.

was larger than that of the control groups. This discrepancy may be attributed to the limited promotional effects of NBW, where the nanobubbles generate •OH radicals upon collapse (Chuenchart et al., 2021; Zhu et al., 2024). It can promote effective sludge solubilization and further degradation of cellular materials in small sludge particles (Yang et al., 2019) while having almost no effects on larger sludge particles. The hydrolysis performance under thermophilic conditions is superior to that under mesophilic conditions (Liu et al., 2013), resulting in a greater number of small particles in thermophilic environments compared with mesophilic ones. A specific quantity of NBW can consume a certain number of small particles, leading to a higher residual count of small particles under thermophilic conditions than under mesophilic conditions. Temperature conditions made a large difference in sludge particle size compared with NBW, thereby representing a more critical factor in influencing sludge particle size.

VFA is the product of hydrolysis in AD, SCOD represents the content of soluble reductive matter, and the trends of SCOD and VFA are similar (Shi et al., 2023; Wang et al., 2023). The SCOD and VFA were higher in the T-M groups than in the T-T groups on day 6 (Fig. 2 **and Supplementary Material**). This observation may be attributed to the addition of mesophilic seed sludge and changing the temperature conditions from thermophilic to mesophilic, the microorganisms in the T-M groups had not fully adapted to the new ambient temperature, and the ability to degrade organic matter had not yet been restored (Postawa et al., 2020). The SCOD of the T-T groups was at a high level, while the VFA was shallow at the late stage of AD. The thermophilic conditions that promoted the degradation of VFA and led to the accumulation of some refractory organics (heterocyclic-N compounds, e.g., pyrimidine, pyrrole, and pyridine) to increase the SCOD (Zhang et al., 2024).

3.2. *Three-dimensional fluorescence excitation and emission matrix spectra*

In the 3D-EEM spectra, regions I and II are simple aromatic proteins; region III is fulvic acids; region IV is soluble microbial by-products; and region V is humic acids (Guan et al., 2024). The fluorescence intensity was maximized in region V, while it was nearly zero in the other regions (Fig. 3). In terms of temperature conditions, the fluorescence intensity of region V was M−M (148.3–149.1) > T-M (62.4–66.7) > T-T (9.0–11.7) on day 41. Humic acids decreased with increasing thermophilic time (Warwick et al., 1997). In addition, adding NBW reduced the fluorescence intensity. Humic acids were an intermediate product of the bioconversion of organic matter to methane during AD and had a complex redox function (Wang et al., 2021). The nanobubbles released chemical energy upon collapse and were accompanied by the production of •OH radicals, which had strong oxidizing properties helping to degrade complex humic acids (Hou et al., 2024). The addition of 25.0 % nanobubbles slightly reduced the fluorescence intensity in region V of aerobic-activated sludge, and NBW accelerated the degradation of humic acids (Xiao et al., 2021). The fluorescence intensities of TTN, TMN, and MMN groups were reduced by 23.1 %, 6.4 %, and 0.5 %, respectively, compared with the control groups without NBW. This might be because the T-T groups contain fewer humic acids, resulting in a greater amplitude of change in fluorescence intensity. HIX indicates the degree of humification of dissolved organic matter (Guan et al., 2024). The trend observed in HIX was consistent with the fluorescence intensity in region V, thereby demonstrating that both NBW and thermophilic conditions facilitated the degradation of humic acids. The increase in humic acids inhibited the ability to produce methane (Li et al., 2023). Consequently, the fluorescence intensity of humic acids exhibited a negative correlation with methane yield, consistent with the results of methane yield (Table 1).

3.3. *Methane yield*

The models were fitted with Gompertz (Table 1), and the fitted results were subjected to Duncan's test. The results showed that the lag phase of the digesters was significantly different under different temperatures ($p < 0.05$). The longest lag phase was observed in the T-M digesters, attributed to the microorganisms needing time to adapt to the new environmental conditions (Postawa et al., 2020). Under theoretical conditions, the R_m of the T-T digesters was the largest and significantly different from the other groups ($p < 0.05$). It has been found that under thermophilic conditions, the larger the R_m, the smaller the λ (Arelli et al., 2021), indicating that the T-T digesters can produce more methane while shortening the reaction period. The peak daily methane yield rate in TTN was higher and appeared earlier than in TTD (Table 1 **and** Fig. 4**a**). NBW accelerated the degradation of the substrate and the

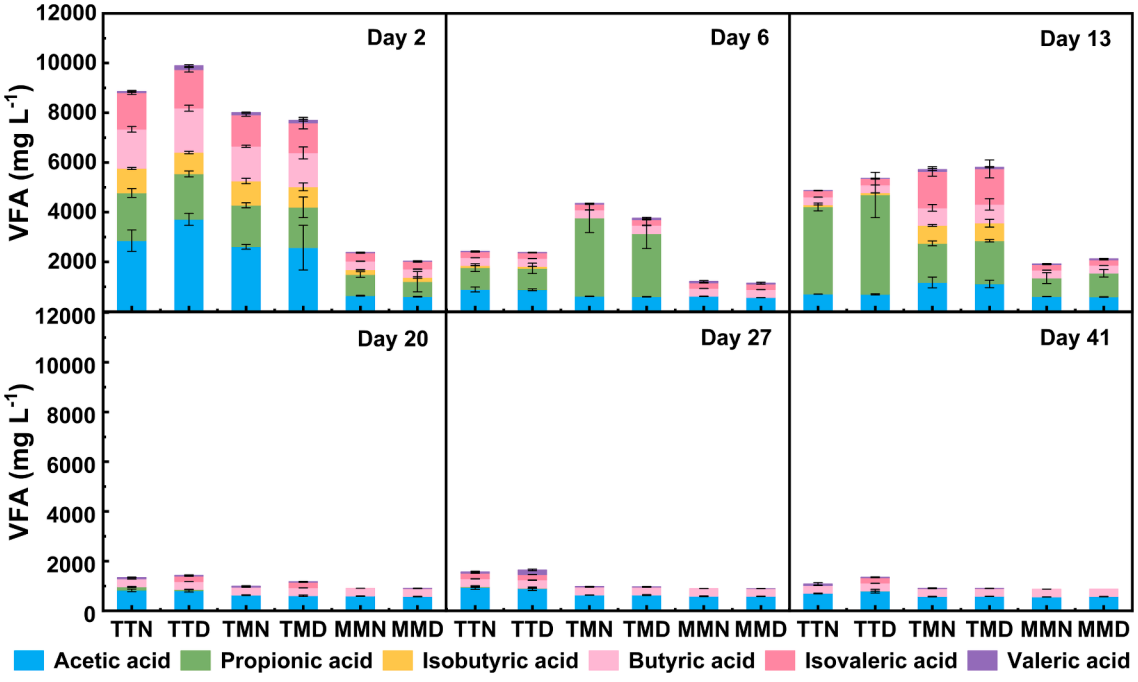


Fig. 2. The volatile fatty acid during anaerobic digestion.

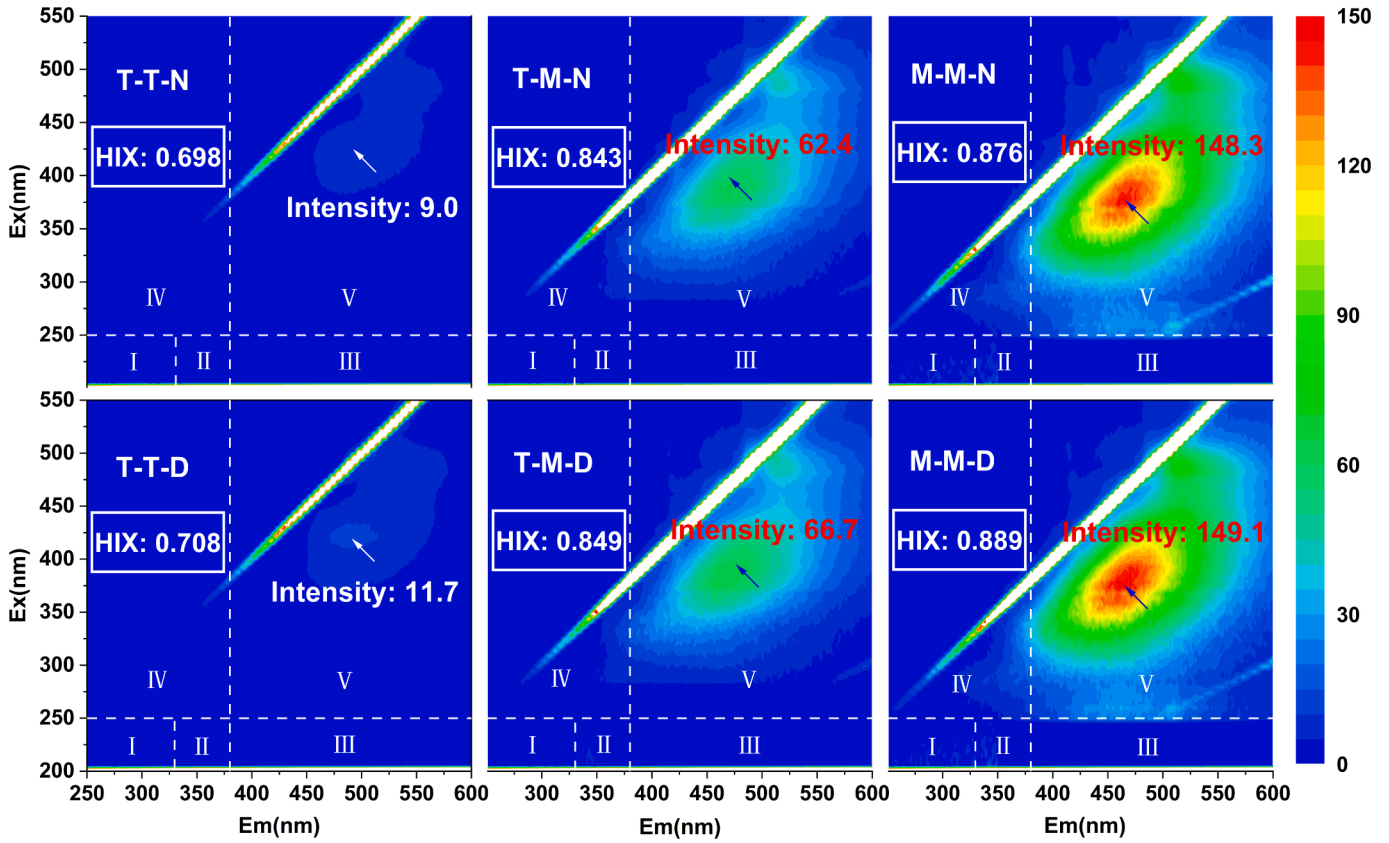


Fig. 3. The 3D-EEM spectra of the biogas slurry on day 41 (the fluorescence intensity based on SCOD of 1000 mg L^{-1}).

Table 1
Parameters of the modified Gompertz model.

	Ultimate methane yield, UMY (mL g^{-1} $VS_{removed}$)	Maximum methane yield rate, R_m (mL g^{-1} $VS_{removed}$ d^{-1})	Lag phase, λ (d)	R^2
TTN	421.88 ± 13.99^{a}	131.98 ± 18.81^{a}	0.90 ± 0.01^{c}	0.995 ± 0.001
TTD	298.13 ± 28.83^{b}	84.12 ± 5.35^{b}	1.01 ± 0.09^{c}	0.996 ± 0.001
TMN	250.58 ± 21.23^{c}	24.29 ± 1.07^{c}	2.30 ± 0.23^{a}	0.982 ± 0.003
TMD	203.30 ± 28.47^{d}	19.86 ± 3.80^{c}	2.23 ± 0.13^{a}	0.971 ± 0.009
MMN	217.92 ± 11.16^{cd}	17.84 ± 0.19^{c}	1.61 ± 0.43^{b}	0.979 ± 0.019
MMD	161.66 ± 25.91^{e}	14.01 ± 2.29^{c}	1.67 ± 0.31^{b}	0.988 ± 0.001

Note: All values are expressed as mean ± standard deviation (n = 3); a, b, c, d, and e were labeled according to Duncan's test. When there were no duplicate labeling symbols between the two groups, they were considered to be significantly different at the $p < 0.05$ level.

reduction of λ (Hou et al., 2021).

On day 41, the T-M digesters enhanced the UMY by 15.0 %-25.8 % ($p < 0.05$) compared with the M−M digesters (Table 1 **and** Fig. 4**c**). Thermophilic conditions are more effective in promoting the hydrolysis of the complex substrate (Wang et al., 2024b). The T-T digesters enhanced the UMY by 46.6 %-68.4 % ($p < 0.001$) compared with the T-M digesters, which suggested that the methanogenesis phase of the AD process in this study is more suitable for thermophilic conditions. However, mesophilic conditions can promote the activity of methanogens during the methanogenesis phase to ensure the stability of methane yield (Chen et al., 2021b). The reason for this difference could be the low VS/TS of DWS (26.4 %) (Table 1), which was lower than that (74.0 %) of Chen et al. (2021b). Temperature can affect the fluidity and mass transfer efficiency of digestate, and substrate with low organic matter content cannot be degraded as expected under mesophilic conditions (Rodríguez-Jiménez et al., 2022). Thermophilic conditions can improve the efficiency of energy transfer and accelerate the consumption of soluble nutrients, leading to more complete hydrolysis and organic matter utilization than mesophilic conditions without worrying about VFA accumulation (Fig. 2), thereby improving methane yield (Jomnonkhaow et al., 2024). Compared with mesophilic conditions, there was a 21.9 %-35.8 % increase in methane yield when the VS/TS of the substrate was 34.2 % under thermophilic conditions (Zhang et al., 2021).

The UMY of the NBW-added groups was significantly enhanced by 23.3 %-41.5 % ($p < 0.05$) compared with the control groups, respectively (Fig. 4**c**). This finding aligns with the results of Wang et al. (2024a), which reported that Air-NBW improved the methane yield of DWS by 27.9 %. NBW can significantly improve methane yield by promoting enzyme activity (hydrolase and coenzyme F420) and increasing electron transfer (Chuenchart et al., 2021). Among them, NBW promoted the most methane yield in the T-T groups, with 41.5 % ($p < 0001$). Under thermophilic conditions, the mass transfer efficiency of digestate was higher (Rodríguez-Jiménez et al., 2022). The sludge particle size was the smallest in the T-T groups, and a large number of charged ions on the surface of the nanobubbles would release chemical energy to produce •OH radicals, which promoted the further degradation of organic matter in small particle sludge by microorganisms (Yang et al., 2019). TTN increased UMY by 161.0 % ($p < 0.001$) compared with MMD. The coupling of thermophilic-thermophilic TPAD with NBW represented one of the optimal strategies to improve the methane yield of DWS with low VS/TS.

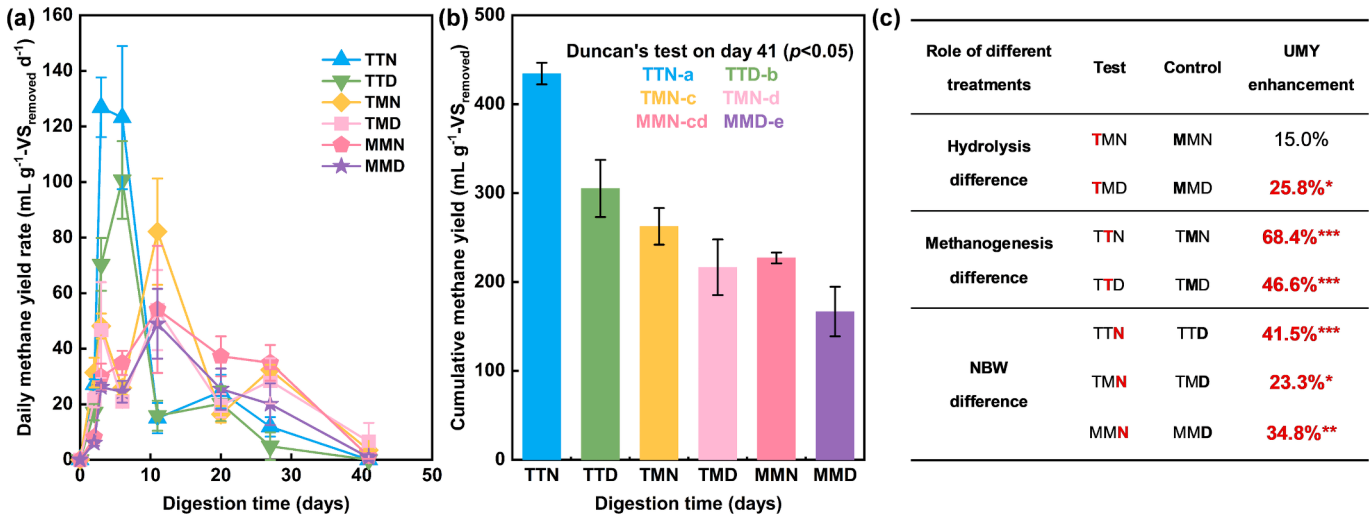


Fig. 4. The (a) daily methane yield rate, (b) cumulative methane yield, and (c) ultimate methane yield enhancement ratio with different treatments during anaerobic digestion.

3.4. Bacterial community

The Chao is proportional to species richness, and the Shannon is proportional to species diversity (Öztep et al., 2023). NBW promoted the diversity and richness of the bacterial community during the hydrolysis phase (Fig. 5**a**). This finding aligns with the results reported by Wang et al. (2020), which indicate that the microaerobic environment created by Air-NBW facilitated direct interspecific electron transfer (DIET) and improved the diversity and richness of microorganisms during the hydrolysis phase.

Firmicutes, *Proteobacteria*, *Actinobacteriota*, *Chloroflexi*, and *Bacteroidota*, were the top five dominant bacteria during AD (Fig. 5**b**). Compared with mesophilic conditions, the relative abundance (RA) of *Firmicutes* (6.5 %-46.0 %) and *Actinobacteriota* (0.9 %-4.0 %) increased, and the RA of *Proteobacteria* (1.8 %-14.8 %), *Chloroflexi* (1.4 %-6.4 %) and *Bacteroidota* (3.9 %-16.7 %) decreased under thermophilic conditions (**see Supplementary Material**). Furthermore, the RA of *Firmicutes* (0.0 %-8.7 %) increased, and the RA of *Proteobacteria* (0.1 %-13.1 %) decreased because of the addition of NBW. PCA showed significant differences between the groups under different temperature conditions (**see Supplementary Material**). There were no significant differences between the NBW-added groups and the control groups. This implied the changes in the bacterial community were mainly controlled by temperature.

Both *Firmicutes* and *Actinobacteriota* were significantly and positively correlated ($p < 0.01$) with SCOD, TAN, UMY, and VFA (Fig. 5**c**). *Firmicutes* can break down complex carbohydrates and proteins to produce VFA, which is converted to H_2 for utilization by hydrogenotrophic methanogens (Ali et al., 2024; Öztep et al., 2023). *Firmicutes* were able to survive both thermophilic and mesophilic conditions. However, the promotion of hydrolysis increased SCOD and VFA under thermophilic conditions, which elevated the RA of *Firmicutes*. Furthermore, NBW increased the activity of hydrolytic enzymes, which further increased the RA of *Firmicutes* (Fan et al., 2021). Similarly, thermophilic conditions elevated the RA of *Actinobacteriota,* and a large amount of organic matter was degraded and converted to acetate and H_2 (de Sousa e Silva et al., 2024). *Proteobacteria* significantly and negatively correlated ($p < 0.01$) with SCOD and VFA. It has been shown that the RA of *Proteobacteria* decreased with increasing hydrolysis under thermophilic conditions (Peng et al., 2024). Thus, *Firmicutes* dominated under thermophilic conditions, whereas *Proteobacteria* dominated under mesophilic conditions. In addition, enhanced hydrolysis by adding NBW further decreased the RA of *Proteobacteria* (Fan et al., 2021).

3.5. Archaeal community

The number of bacteria was much greater than that of archaea, indicating that bacteria dominated the degradation of organic matter while archaea played a supplementary role in degrading organic matter (Fan et al., 2022). NBW promoted the diversity and richness of the archaeal community during the hydrolysis phase (Fig. 6**a**).

Methanothermobacter, *Methanosaeta*, *Methanosarcina*, *Methanobacterium,* and *Methanospirillum* were the top five dominant archaea during AD (Fig. 6**b**). Compared with mesophilic conditions, the RA of *Methanothermobacter* (0.1 %-31.6 %) and *Methanosarcina* (0.1 %-4.2 %) increased, and the RA of *Methanosaeta* (7.9 %-14.8 %), *Methanobacterium* (4.2 %-8.6 %), and *Methanospirillum* (0.5 %-1.2 %) decreased during both the hydrolysis and methanogenesis phases under thermophilic conditions (**see Supplementary Material**). Furthermore, NBW increased the RA of *Methanosarcina* (0.2 %-4.4 %) and *Methanospirillum* (0.0 %-0.6 %) during the hydrolysis phase, decreased the RA of *Methanosarcina* (1.8 %-3.5 %) and *Methanospirillum* (0.0 %-1.1 %) during the methanogenesis phase. PCA showed significant differences between the groups under different temperature conditions (**see Supplementary Material**). There were no significant differences between the NBW-added groups and the control groups. This meant that, like the bacterial community, changes in the archaeal community were primarily controlled by temperature.

The hydrogenotrophic methanogens *Methanothermobacter* were significantly and positively correlated ($p < 0.01$) with SCOD, TAN, UMY, and VFA (Fig. 6**c**). Elevated levels of ammonia or VFA can lead to the enrichment of *Methanothermobacter* (Hori et al., 2006; Yan et al., 2020). During the hydrolysis phase, thermophilic conditions promoted the production of large amounts of VFA and increased the TAN. Subsequently, during the methanogenesis phase, thermophilic conditions promoted the degradation of organic matter, resulting in a continued increase in TAN. Additionally, *Methanothermobacter* has been shown to enhance methane generation under ammonia inhibition (Yan et al., 2020). Consequently, *Methanothermobacter* had an elevated RA.

Mixed trophic methanogens *Methanosarcina* were significantly and positively correlated ($p < 0.01$) with pH, TAN, FAN, and UMY. *Methanosarcina* could be stabilized under conditions of high ammonia levels and high pH levels (Wang et al., 2024b). Thermophilic conditions promoted hydrolysis, generating large amounts of ammonia nitrogen, leading to elevated RA of *Methanosarcina*. Furthermore, NBW increased the RA of *Methanosarcina* during the hydrolysis phase and decreased the RA of *Methanosarcina* during the methanogenesis phase. This was

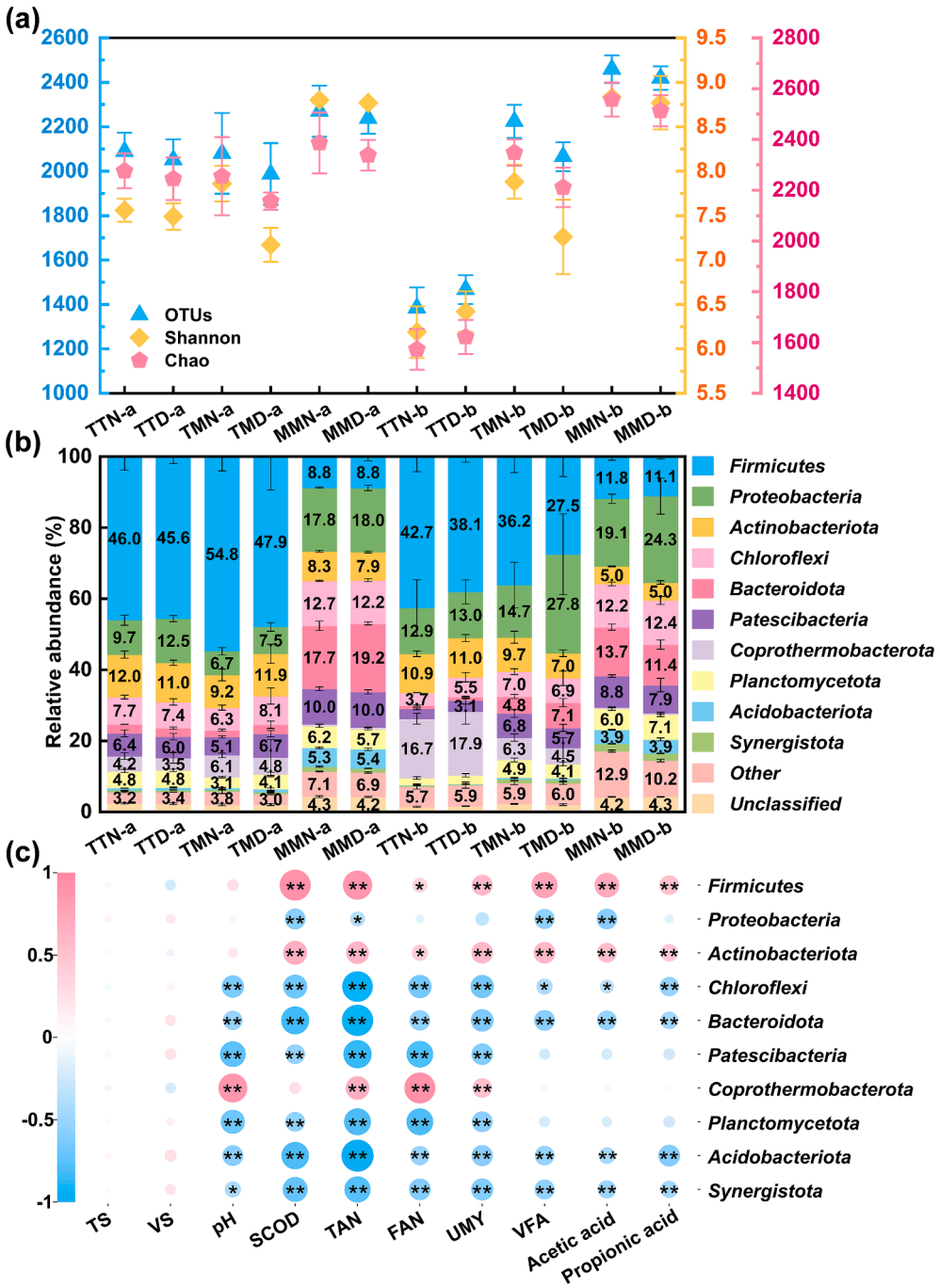


Fig. 5. The (a) Alpha diversity analysis and (b) phylum-level relative abundance of bacteria (a: day 2; b: day 41), and (c) correlation analysis between environmental factors and bacteria.

attributed to the fact that *Methanosarcina* is a facultative bacteria, and the microaerobic environment created by NBW enhanced the activity of *Methanosarcina* and accelerated the hydrolysis (Chuenchart et al., 2021). However, in the methanogenesis phase, where the effects of NBW was weakened, the disappearance of the microaerobic environment reduced the activity of the facultative bacteria, causing a decrease in the RA of *Methanosarcina*.

In the T-T groups, the archaeal community was dominated by hydrogenotrophic methanogens. The H_2 produced by *Firmicutes* and *Actinobacteriota* was utilized by *Methanothermobacter* to increase methane yield. *Firmicutes* controlled the limited rate step of the AD, and thermophilic conditions promoted the hydrolysis of *Firmicutes*. NBW further enhanced the activity of *Firmicutes* and alleviated acid inhibition and ammonia inhibition (Hou et al., 2024). In addition to the role of microorganisms, the •OH radicals generated at the gas–liquid interface during the collapse of nanobubbles, along with elevated temperatures, can disrupt the stable structure of the complex substrate (Hou et al., 2021; Liu et al., 2013). Collectively, these factors contributed to the improvement of AD performance.

All index results showed that the combined application of NBW and MEC did not inhibit or affect each other. The negative surface charge of NBW can increase the amount of electrons available at the electrode (Hou et al., 2024). MEC enriched electroactive microorganisms and methanogens on the electrode surface (Wang et al., 2022a), and the addition of NBW stimulated both microbial activity and electron transfer on the electrode surface, thereby activating the DIET mechanism within

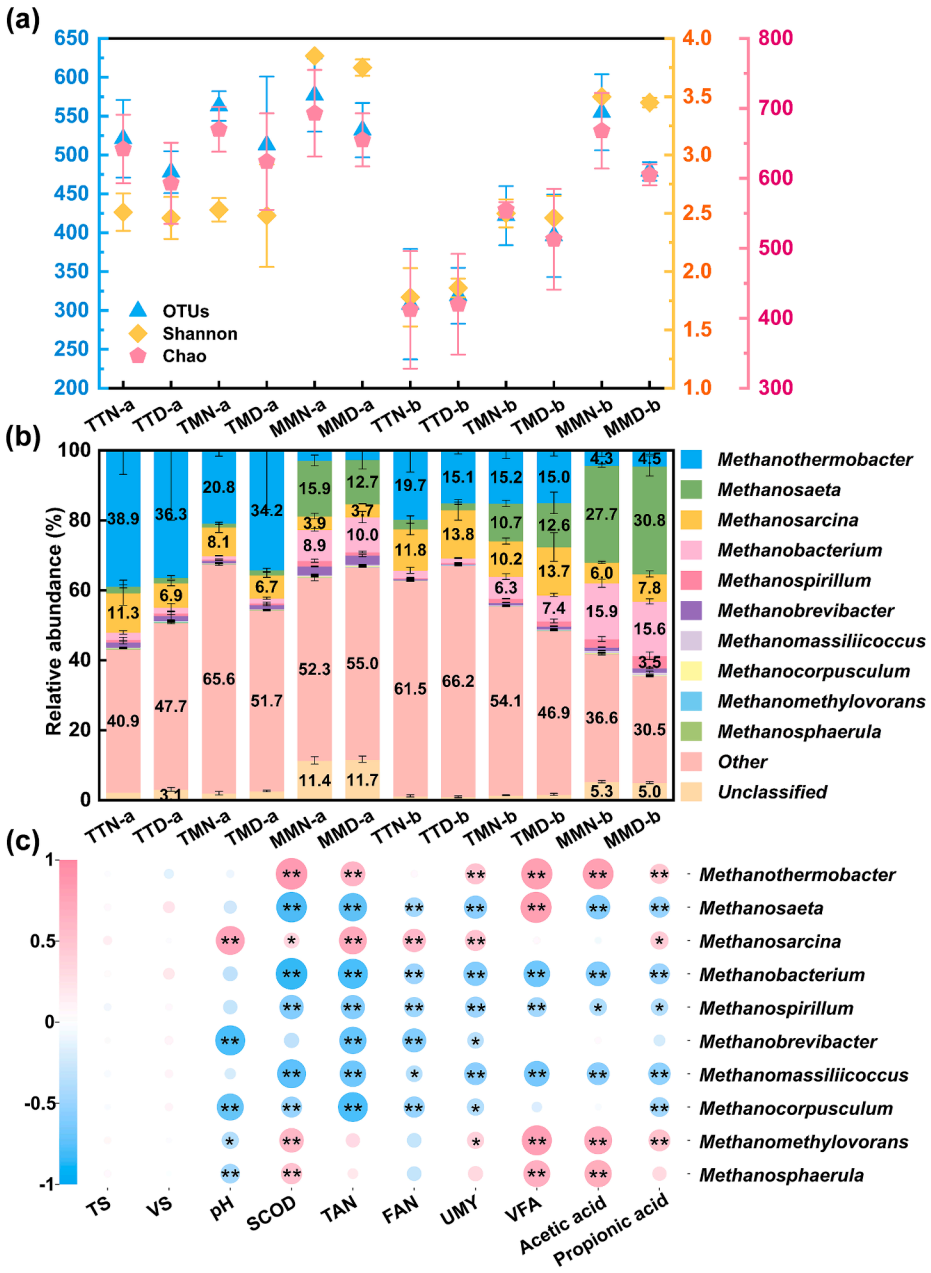


Fig. 6. The (a) Alpha diversity analysis and (b) genus-level relative abundance of archaea (a: day 2; b: day 41), and (c) correlation analysis between environmental factors and archaea.

the AD system and accelerating the electron transfer between microorganisms (Liu et al., 2024). However, the specific change mechanisms underlying the combined effects of NBW and MEC on enzyme activity and metabolic pathways remain unclear, necessitating further investigation.

The combined application of NBW with MEC represents a promising approach. However, due to the instability of sludge organic matter, the effects of different temperature conditions on sludge AD may fluctuate. In the future, the combination of the application of MEC and nanobubble technology could be explored to address additional challenges, such as improving dehydration performance and reducing heavy metal content.

4. Conclusions

NBW synchronized with MEC-TPAD improved hydrolysis and methanogenesis of the DWS digestion process. Among them, the combined application of NBW and thermophilic-thermophilic TPAD is beneficial for AD of sludge with lower organic matter content. NBW significantly promoted hydrolysis under mesophilic-mesophilic conditions, with TAN increased by 13.7 %. NBW significantly promoted methanogenesis under thermophilic-thermophilic conditions, with UMY increased by 41.5 %. NBW increased the diversity and richness of microorganisms in the hydrolysis phase, and the enriched functional microorganisms played an important role in improving AD performance. This research presents a new technological insight to improve the AD performance of DWS.

CRediT authorship contribution statement

Ziying Xu: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Tianfeng Wang:** Writing – review & editing, Supervision, Resources,

Project administration, Methodology, Investigation, Funding acquisition. **Cheng Peng:** Validation, Investigation, Formal analysis. **Yutong Feng:** Investigation, Formal analysis. **Xin Fan:** Investigation, Formal analysis. **Xuan Yang:** Investigation, Formal analysis. **Wenqi Gao:** Resources, Investigation. **Qingfang Zhang:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2025.132084.

Data availability

Data will be made available on request.

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